RMO(INDIA) – 2012

Max Mark – 102 Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.
- All Questions carry equal marks.
- Answer to each questions starts on a new page. Clearly indicate the question number.

1. Let ABC be a triangle. Let D be a point on segment BC such that DC = 2BD. Let E be the mid point of AC. Let AD and BE intersect in P. Determine the ratio $\frac{BP}{PE}$ and $\frac{AP}{PD}$.
2. Let a, b and c are positive integers such that a divides b^3, b divides c^3 and c divides c^3**.** Prove that abc divides $(a+b+c)^{13}$.
3. Let a and b are positive real numbers such that $a+b=1$. Prove that $a^a b^b + a^b b^a \leq 1$.
4. Let $X = \{1,2,3,4,5,6,7,8,9,10\}$. Find the number of pairs $\{A, B\}$ such that $A \subseteq B, B \subseteq X, A \neq B$ and $A \cap B = \{2,3,5,7\}$
5. Let ABC be a triangle and let BE, CF be respectively internal angle bisector of $\angle B, \angle C$ with E on AC and F on AB. Let X is a point on the segment CF such that $AX \perp CF$ and Y is a point on the segment BE such that $AY \perp BE$. Prove that $XY = (b+c-a)/2$, where $a = BC, b = CA, c = AB$.
6. Let a and b are real numbers such that $a \neq 0$. Prove that not all the roots of $ax^4 + bx^3 + x^2 + x + 1 = 0$ can be real.

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